

RUTHVIKA TANGIRALA

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SUMMARY

Cloud Data Engineer with **4+ years of experience** building and maintaining data pipelines across **mobility, automotive, enterprise technology, and energy sectors**. Experienced in developing **batch ingestion, CDC, and streaming pipelines** on **AWS (S3, Glue, Redshift, Kinesis, Airflow)** to process large structured and semi-structured datasets for analytics and machine learning workloads. Skilled in **dimensional modeling, lakehouse architectures, and distributed processing** using **Apache Spark, Kafka, and Snowflake**, with additional experience on **Azure data platforms**. Familiar with **GenAI data workflows**, including **RAG pipelines, vector databases, and LangChain**.

PROFESSIONAL EXPERIENCE

Cloud Data Engineer (AWS)

USA

Uber Technologies Inc.

October 2025 – Present

- Created batch ingestion pipelines using **AWS Glue, Amazon S3, and Apache Spark** to process **8+ terabytes** of structured and semi-structured rider and trip data into **curated analytics layers** used by **machine learning** and marketplace analytics systems.
- Built **Change Data Capture (CDC) workflows** to incrementally process booking, dispatch, and trip status updates from source systems into analytical data stores, reducing latency from **3 hours to under 45 minutes** between operational events and reporting availability.
- Implemented **automated data quality checks** and monitoring for **freshness, schema drift, and volume anomalies** across **5+ production datasets**, reducing downstream reporting incidents by **47%**.
- Optimized **Spark SQL transformations** through **partition pruning, join refactoring, and shuffle reduction**, cutting production pipeline runtime by **41%** and lowering compute costs by **30% annually**.
- Established **pipeline orchestration and monitoring** using **Apache Airflow** and **AWS CloudWatch** across **10+ production pipelines**, enabling automated retries and data validation checks for consistent daily data availability.

Data Engineer

USA

Mercedes-Benz R&D

September 2024 – May 2025

- Architected scalable **ELT pipelines** in **AWS (S3, Glue, Redshift)** to ingest and process **3M+ connected vehicle telemetry records**, enabling refreshed vehicle performance dashboards every 3 hours instead of next-day batch reporting.
- Modeled **dimensional schemas** in **Amazon Redshift** consolidating 14 vehicle subsystem, sensor, and diagnostic data sources into governed fact tables supporting analytics for 120+ engineering and product stakeholders.
- Investigated telemetry datasets using **advanced SQL** (CTEs, window functions, multi-table joins) to uncover recurring performance anomalies, contributing to early detection of fault patterns during a 4-month fleet monitoring cycle.
- Evaluated statistical trends in **Python (Pandas, SciPy)** on vehicle usage and component stress metrics, providing insights that informed design adjustments across three high-variance performance segments.
- Partnered directly with engineering managers and product leads to translate diagnostic reporting requirements into **Tableau dashboards** tracking 20+ vehicle health KPIs, eliminating **60+ hours** of monthly manual data consolidation.

Data Analyst

India

IBM

February 2022 – August 2023

- Developed interactive dashboards in Power BI using DAX to track 25+ supply chain KPIs, consolidating fragmented Excel-based reports and reducing executive reporting preparation time from 4 days to 2 days.
- Constructed **ETL pipelines** in **Azure Data Factory** integrating data from five enterprise systems into **Azure Synapse Analytics**, reducing batch refresh cycles from 2 hours to 70 minutes.
- Structured **dimensional data models** in **SQL Server** using **Star Schema** principles to organize **7M+ transactional records** for scalable analytical querying.
- Instituted **data lineage and metadata governance** within Azure Synapse and **Power BI semantic layers**, strengthening auditability and enforcing **role-based access control (RBAC)** across enterprise datasets.
- Streamlined data extraction and transformation workflows using **Python (Pandas)** and VBA, replacing seven manual Excel processes and saving **80 analyst hours per month**.
- Ingested streaming transaction logs from **Apache Kafka** into **Azure Data Lake**, supporting **900K daily records** and enabling near real-time anomaly monitoring.
- Collaborated within Agile (Scrum) teams to translate BRD/FRD requirements into analytical data models and calculated measures, contributing to delivery of 4 BI initiatives within planned sprint timelines.

- Assessed sensor datasets using **Python** and **SQL** to identify energy usage inefficiencies, contributing to workflow improvements with potential **\$1.2M annual savings**.
- Delivered a **Tableau dashboard** monitoring energy consumption trends across multiple plants, highlighting areas of excess consumption.
- Deployed **automated SQL validation scripts** across **40+ datasets**, improving data consistency between sensor and billing systems for accurate financial reporting.
- Optimized **Snowflake pipelines** with incremental processing logic and applied **PySpark-based anomaly detection** on transaction samples to help prevent potential revenue loss.

TECHNICAL SKILLS

Programming & Data Processing: Python (Pandas, NumPy, PySpark), SQL (Advanced Query Optimization, CTEs, Window Functions), Spark SQL, R, KQL (Kusto Query Language), Bash, PowerShell

Cloud Data Engineering (AWS): Amazon S3, AWS Glue, Amazon Redshift, Amazon Athena, AWS Lambda, Amazon EC2, AWS EMR, AWS Kinesis Data Streams, AWS QuickSight, AWS EventBridge

Data Architecture & Modeling: Lakehouse Architecture, Medallion Architecture (Bronze/Silver/Gold), Dimensional Modeling, Star Schema, Snowflake Schema, Data Contracts, Change Data Capture (CDC), Incremental Data Processing, Metadata Management

Big Data & Distributed Processing: Apache Spark, Databricks, Delta Lake, Distributed Data Processing, Partitioning Strategies, Columnar Storage Optimization

Streaming & Event-Driven Systems: Apache Kafka, AWS Kinesis, Event-Driven Architecture, Streaming Data Pipelines, Real-Time Data Processing

Data Warehousing & Databases: SQL Server, PostgreSQL, MySQL, Oracle, Amazon Redshift, Snowflake, Google BigQuery

Data Integration & Orchestration : Apache Airflow, Azure Data Factory, CI/CD Data Pipelines, Workflow Automation, Data Pipeline Scheduling

Business Intelligence & Analytics Engineering: Power BI (DAX, Semantic Models, DirectLake), Tableau, Looker, AWS QuickSight, KPI Modeling, Executive Data Visualization

DevOps & Infrastructure for Data Platforms: Git, GitHub, Docker, Kubernetes, Azure DevOps, GitHub Actions, CI/CD Pipelines, Agile (Scrum), Jira

Security, Governance & Data Reliability: Role-Based Access Control (RBAC), Row-Level Security (RLS), Data Lineage, Audit Logging, Compliance Monitoring, Microsoft Purview

AI & Machine Learning Data Enablement: Feature Engineering Pipelines, ML Training Dataset Preparation, Model-Ready Data Pipelines, Vector Databases, Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, MLflow, Generative AI Data Workflows

PROJECTS

Traffic Stops Analysis & Visualization

- Built a serverless AWS pipeline (Lambda, EventBridge, Glue, S3, Redshift) processing **1.2M+ traffic records**, reducing preprocessing time **6 hrs to 40 mins** and enabling **QuickSight KPI dashboards**.

Stock Market & News Sentiment Analytics

- Built a cloud-based **OLAP** warehouse on AWS (S3, Glue PySpark, Athena), transforming **40,000+** stock and sentiment records into a **Star Schema** and cutting average query runtime by **59% (22s to 9s)** via partitioning and column pruning.

Personal Trainer AI Agent (GenAI Project)

- Built a **RAG AI system** using **LangChain**, **Pinecone**, and **GCP**, indexing **3K+ documents** with **MLflow**, **Docker**, **Airflow**, and **GitHub Actions** for automated deployment.

EDUCATION

Northeastern University | USA

Master of Science, Data Science and Analytics

September 2023 – July 2025

CERTIFICATIONS

[AWS Certified Cloud Practitioner](#) - Amazon Web Services (2022)

[Databases and SQL for Data Science](#) - Coursera (2020)

[Data Analytics with Python](#) - NPTEL (2021)

[Python for Data Science, AI & Development](#) - Coursera (2020)

[Data Cleaning & Data Manipulation](#) - Kaggle (2024)